

homogenous linear systems

before we actually begin we will do a review on matrices

$$\underline{x'} = Ax$$

homogenous
Form

$$\underline{x'} = Ax + f$$

non-homogenous
Form

$$\begin{matrix} & C_1 & C_2 & C_3 \\ \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} & \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \end{matrix}$$

$R \times C$
row column

this is a 3×3 matrix

$$\begin{matrix} & C_1 & C_2 & C_3 \\ \begin{matrix} R_1 \\ R_2 \end{matrix} & \begin{bmatrix} a & b & c \\ e & f & g \end{bmatrix} \end{matrix}$$

this is a 2×3 matrix

you always begin with row then the column so:

row $\times$ column

★

Dependent Variable: $x_1, x_2, x_3, \dots, x_n$

independent Variable: t

• if you have "t" in your problem you are dealing with a independent

how to write into matrix form:

$$\frac{dx}{dt} = 6x + y + z + t$$

$$\frac{dy}{dt} = 8x + 7y - z + 10t$$

$$\frac{dz}{dt} = 2x + 9y - z + 6t$$

$$x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad x' = \begin{bmatrix} 6 & 1 & 1 \\ 8 & 7 & -1 \\ 2 & 9 & -1 \end{bmatrix} x, \quad f = \begin{bmatrix} 1 \\ 10 \\ 6 \end{bmatrix} t$$

$$x' = \begin{bmatrix} 6 & 1 & 1 \\ 8 & 7 & -1 \\ 2 & 9 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 10 \\ 6 \end{bmatrix} t$$

Multiplication with matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}, \quad C = \begin{bmatrix} m \\ n \end{bmatrix}$$

$$a) A \cdot B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

$$b) A \cdot C = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} m \\ n \end{bmatrix} = \begin{bmatrix} am + bn \\ cm + dn \end{bmatrix}$$

you will be dealing with three cases:

1) two different values

2) repeating values

3) complex conjuncts

this is an equation you will need to know for all cases:

$$\det(A - \lambda I)$$

↑
Determinant

First case: two different numbers

$$\frac{dx}{dt} = 2x + 3y$$

$$\frac{dy}{dt} = 6x + 5y$$

$$\det(A - \lambda I) = \begin{vmatrix} 2 & 3 \\ 6 & 5 \end{vmatrix}$$

• all we are doing here is taking the numbers from the equations and plugging them into a matrix

We will plug the diagonal numbers for "A" in the equation

$$\begin{vmatrix} 2-\lambda & 3 \\ 6 & 5-\lambda \end{vmatrix}$$

★ you only do it for the diagonal

Characteristic equation

$$(2-\lambda)(5-\lambda) - (3)(6)$$

$$10 - 2\lambda - 5\lambda + \lambda^2 - 18$$

$$\lambda^2 - 7\lambda - 8$$

$$(\lambda - 8)(\lambda + 1)$$

$$\lambda_1 = 8 \quad \lambda_2 = -1$$

• we get two different values

eigenvalues

@ $\lambda_1 = 8$

Plug in $\lambda = 8$ into the matrix

$$\begin{array}{c} R_1 \downarrow \\ \begin{vmatrix} -6 & 3 \\ 6 & -3 \end{vmatrix} \\ R_2 \uparrow \end{array} \quad \begin{array}{c} \downarrow R_1 \\ \begin{vmatrix} 3 \\ -3 \end{vmatrix} \\ \uparrow R_2 \end{array}$$

• our next step to to create an equation that will get the second row to zero

$R_2 + R_1$ • this is our equation we can test it:

$$(-6) + (6) = 0$$

$$(3) + (-3) = 0$$

$$\begin{vmatrix} -6 & 3 \\ 0 & 0 \end{vmatrix}$$

• we want to multiply this by something that will make this one the top row stays the same

$$-\frac{1}{6} R_1 \longrightarrow \begin{vmatrix} x_1 & x_2 \\ 1 & -\frac{1}{2} \\ 0 & 0 \end{vmatrix}$$

• we take the first row and do this to it:

$$x_1 - \frac{1}{2} x_2 = 0$$

$$x_1 = \frac{1}{2} x_2$$

now we need to choose a number for x_1 and x_2 and plug it in that will get both sides equal to one another

$$x_1 = \frac{1}{2} x_2$$

$$(1) = \frac{1}{2} (2)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

we still need to do $\lambda_2 = -1$ we will do the same exact thing we did for $\lambda_1 = 8$

@ $\lambda = -1$

$$\begin{array}{ccc} \begin{array}{c} \downarrow \\ \left| \begin{array}{cc|c} 3 & 3 & 0 \\ 6 & 6 & 0 \end{array} \right| \end{array} & \xrightarrow{2R_1 - R_2} & \begin{array}{c} \left| \begin{array}{cc|c} 3 & 3 & 0 \\ 0 & 0 & 0 \end{array} \right| \end{array} \end{array} \quad \xrightarrow{\frac{1}{3} R_1}$$

$$\begin{array}{cc|c} x_1 & x_2 & \\ \hline 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}$$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

$$(-1) = -(1)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$x = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t}$$

when you are writing your final equation for case #1

$$x = c_1 \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} e^{\lambda_1 t} + c_2 \begin{pmatrix} u_3 \\ u_4 \end{pmatrix} e^{\lambda_2 t}$$

Second case: repeating factors

$$x' = \begin{pmatrix} -1 & 5 \\ -5 & 9 \end{pmatrix} x$$

• Second case problems don't have to always look like this

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & 5 \\ -5 & 9-\lambda \end{vmatrix}$$

• you only do this diagonally

$$(-1-\lambda)(9-\lambda) - (-5)(5)$$

$$-9 + \lambda - 9\lambda + \lambda^2 + 25$$

$$\lambda^2 - 8\lambda + 16$$

$$(\lambda - 4)(\lambda - 4)$$

$$\lambda_1 = 4, \lambda_2 = 4$$

@ $\lambda_1 = 4$

• Plug this in the " λ " in the matrix

$$\begin{vmatrix} -5 & 5 & 0 \\ -5 & 5 & 0 \end{vmatrix}$$

• when you are dealing with case #2 you need to include this

• we need to find an equation that will get the second row to zero:

$$-R_1 + R_2$$

We can test the equation we created:

$$-(-5) + (-5) = 0$$

$$-(5) + (5) = 0$$

$$\left| \begin{array}{cc|c} -5 & 5 & 0 \\ 0 & 0 & 0 \end{array} \right|$$

• we want to multiply this by something to get it equal to one

$$\xrightarrow{-1/5 R_1} \begin{array}{cc|c} x_1 & x_2 & \\ \hline 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}$$

• now we take the first row and do this with it:

$$x_1 - x_2 = 0$$

$$x_1 = x_2$$

We now choose any number for x_1 and x_2 to get both sides equal to each other

$$x_1 = x_2$$

$$(1) = (1)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

first equation:

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t}$$

We now need to find the second equation because we have repeating terms we will have the same matrix just with one small difference

$$\left[\begin{array}{cc|c} -5 & 5 & 1 \\ -5 & 5 & 1 \end{array} \right]$$

• these are from first equation

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t}$$

we now want to create an equation to get the and row equal to zero

$$-R_1 + R_2 \rightarrow \left[\begin{array}{cc|c} -5 & 5 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

• we now want to get this equal to one

• we also apply the equation here

$$-1/5 R_1 \rightarrow \begin{array}{cc} P_1 & P_2 \\ \left[\begin{array}{cc|c} 1 & -1 & 1/5 \\ 0 & 0 & 0 \end{array} \right] \end{array}$$

• we now do this to the first row:

$$P_1 - P_2 = 1/5$$

we can set P_2 equal to anything we will set $P_2 = 1$

$$P_1 - (1) = 1/5$$

$$P_1 = 4/5$$

$$P_1 = 4/5$$

$$P_2 = 1$$

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} 4/5 \\ 1 \end{pmatrix}$$

second equation:

$$x_2 = v t e^{\lambda_1 t} + p e^{\lambda_2 t}$$

• this is from equation one

$$x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} t e^{4t} + \begin{pmatrix} 4/5 \\ 1 \end{pmatrix} e^{4t}$$

$$x = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t} + c_2 \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t e^{4t} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{4t} \right]$$

the last one we are left with is case #3

case #3: complex conjugates

$$\frac{dx}{dt} = 6x + y$$

$$\frac{dy}{dt} = -2x + 4y$$

$$\det(A - \lambda I) = \begin{vmatrix} 6-\lambda & 1 \\ -2 & 4-\lambda \end{vmatrix}$$

• you only do the diagonal ones

$$\begin{aligned} (6-\lambda)(4-\lambda) - (-2)(1) \\ 24 - 6\lambda - 4\lambda + \lambda^2 + 2 \\ \lambda^2 - 10\lambda + 26 \end{aligned}$$

• notice that you will need the quadratic formula to factor this

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \longrightarrow \frac{-(-10) \pm \sqrt{(10)^2 - 4(1)(26)}}{2(1)} \longrightarrow \frac{10 \pm 2i}{2}$$

$$\Rightarrow 5 \pm i$$

we will only use the positive one when we are dealing with complex conjugates:

$$5 + i$$

we will now take " $s+i$ " and plug it in for " λ " in the matrix:

$$\begin{vmatrix} 6 - (s-i) & 1 \\ -2 & 4 - (s+i) \end{vmatrix} \Rightarrow \begin{vmatrix} 1-i & 1 \\ -2 & -1-i \end{vmatrix}$$

we now need to create an equation to get the second row equal to zero:

$$2R_1 + (1-i)R_2 \rightarrow \begin{bmatrix} x_1 & x_2 \\ 1-i & 1 \\ 0 & 0 \end{bmatrix}$$

• when we are doing complex conjugate we don't need **this** to be one

we now do this with the first row:

$$(1-i)x_1 + x_2 = 0$$

$$(1-i)x_1 = -x_2$$

then we want to get both sides equal to each other by plugging any amount into " x_1 " and " x_2 "

$$(1-i)(-1) = -(1-i)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 \\ 1-i \end{bmatrix} \Rightarrow \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} i$$

• we broke it into real part and imaginary part
this is what you will use to write the final

this is how the final equation should be formatted:

$$x = \left[(B_1 \cos(\beta t) - B_2 \sin(\beta t)) e^{\alpha t} + \right. \\ \left. (B_2 \cos(\beta t) + B_1 \sin(\beta t)) e^{\alpha t} \right]$$

$$x = C_1 \left[\begin{pmatrix} -1 \\ 1 \end{pmatrix} \cos(ct) - \begin{pmatrix} 0 \\ -1 \end{pmatrix} \sin(ct) \right] e^{st} + C_2 \left[\begin{pmatrix} 0 \\ -1 \end{pmatrix} \cos(ct) + \begin{pmatrix} -1 \\ 1 \end{pmatrix} \sin(ct) \right] e^{st}$$